

Module 3 : Decision Theory

Decision Theory : Introduction ,Decision under uncertainty - Maxmin & Minmax, Decision under Risk - Expected Value, Simple decision tree problems.(Only theory).

Job Sequencing: Job Sequencing - N Jobs - two machines and N Jobs - three machines, n jobs on m machines, Sequencing of 2 jobs - M machines cases.(Theory and Problems)

DECISION THEORY

I. MEANING : The decision theory is a technique used for decision making in an uncertain condition or situation. It provides a framework and methodology for rational decision making when the outcomes are uncertain.

Decision making is the thinking process of picking a choice between few of the available options. It serves as a purpose to remove any uncertainty and hesitation before deciding on something.

II. ZONES OF DECISION THEORY

Ignorance	Uncertainty	Risk	Certainty
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Increasing Knowledge 

Decision theory is both descriptive and prescriptive business modelling approach to classify the degree of knowledge and compare expected outcomes due to several courses of action. The degree of knowledge is divided into four categories : *complete knowledge (i.e. certainty), ignorance, risk and uncertainty.*

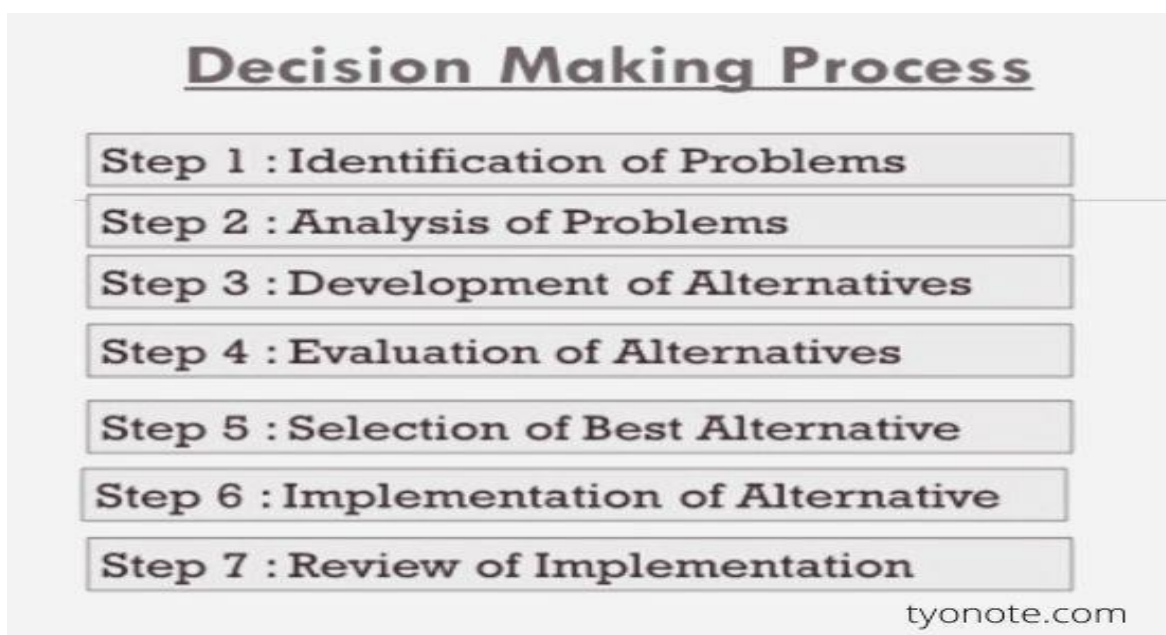
III. COMPONENTS OF DECISION THEORY

- 1. Decision alternatives:** There is a finite number of decision alternatives available to the decision-maker at each point in time when a decision is made. The number and type of such alternatives may depend on the previous decisions made and their outcomes.
- 2. States of nature:** A state of nature is an event or scenario that is not under the control of decision makers. For instance, it may be the state of economy (e.g. inflation), a weather condition, a political development, etc.
- 3. Payoff:** It is a numerical value (outcome) obtained due to the application of each possible combination of decision alternatives and states of nature. The payoff values are always conditional values because of unknown states of nature. The payoff values are measured within a specified period (e.g. within one year, month, etc.) called the *decision horizon*. The payoffs in most decisions are monetary. Payoffs resulting from each possible combination of decision alternatives and states of natures are displayed in a matrix.

IV. STEPS OF DECISION MAKING PROCESS

The decision-making process involves the following steps:

1. Identify and define the problem.
2. List all possible future events (not under the control of decision-maker) that are likely to occur.
3. Identify all the *courses of action* available to the decision-maker.
4. Express the payoffs resulting from each combination of course of action and state of nature.
5. Apply an appropriate decision theory model to select the best course of action from the given list on the basis of a criterion (measure of effectiveness) to get optimal (desired) payoff.



V. TYPES OF DECISION MAKING ENVIRONMENTS

To arrive at an optimal decision it is essential to have an exhaustive list of decision-alternatives, knowledge of decision environment, and use of appropriate quantitative approach for decision-making. In this section three types of decision-making environments: *certainty*, *uncertainty*, and *risk*, have been discussed. The knowledge of these environments helps in choosing the quantitative approach for decision-making.

1. Type 1 Decision-Making under Certainty

In this decision-making environment, decision-maker has complete knowledge (perfect information) of outcome due to each decision-alternative (course of action). In such a case he would select a decision alternative that yields the maximum return (payoff) under known state of nature.

For example, the decision to invest in *National Saving Certificate*, *Indira Vikas Patra*, *Public Provident Fund*, etc., is where complete information about the future return due and the principal at maturity is known.

2. Type 2 Decision-Making under Risk

In this decision-environment, decision-maker does not have perfect knowledge about possible outcome of every decision alternative. It may be due to more than one states of nature. In such a case he makes an assumption of the probability for occurrence of particular state of nature.

3. Type 3 Decision-Making under Uncertainty

In this decision environment, decision-maker is unable to specify the probability for occurrence of particular state of nature. However, this is not the case of decision-making under ignorance, because the possible states of nature are *known*. Thus, decisions under uncertainty are taken even with less information than decisions under risk.

For example, the probability that Mr X will be the prime minister of the country 15 years from now is not known.

VI. DECISION - MAKING UNDER UNCERTAINTY

Decision making under uncertainty is an environment in which the probability of outcomes or states of nature cannot be quantified.

The following criteria of decision-making under uncertainty have been discussed in this section.

1. Optimism (Maximax or Minimin) criterion
2. Pessimism (Maximin or Minimax) criterion
3. Equal probabilities (Laplace) criterion
4. Coefficient of optimism (Hurwicz) criterion
5. Regret (salvage) criterion

1. Optimism (Maximax or Minimin) Criterion : In this criterion the decision-maker ensures that he should not miss the opportunity to achieve the largest possible profit (maximax) or the lowest possible cost (minimin). Thus, he selects the decision alternative that represents the maximum of the maxima (or minimum of the minima) payoffs (consequences or outcomes).

The working method is summarized as follows:

- Locate the maximum (or minimum) payoff values corresponding to each decision alternative.
- Select a decision alternative with best payoff value (maximum for profit and minimum for cost).

Since in this criterion the decision-maker selects an decision-alternative with largest (or lowest) possible payoff value, it is also called an *optimistic decision criterion*.

2. Pessimism (Maximin or Minimax) Criterion : In this criterion the decision-maker ensures that he would earn no less (or pay no more) than some specified amount. Thus, he selects the decision alternative that represents the maximum of the minima (or minimum of the minima in case of loss) payoff in case of profits. The working method is summarized as follows:

- Locate the minimum (or maximum in case of profit) payoff value in case of loss (or cost) data corresponding to each decision alternative.
- Select a decision alternative with the best payoff value (maximum for profit and minimum for loss or cost).

Since in this criterion the decision-maker is conservative about the future and always anticipates the worst possible outcome (minimum for profit and maximum for cost or loss), it is called a *pessimistic decision criterion*. This criterion is also known as *Wald's criterion*.

VII. DECISION - MAKING UNDER RISK

Decision making risk is an environment in which the probability of outcomes or states of nature can be quantified.

In this decision-making environment, decision-maker has sufficient information to assign probability to the likely occurrence of each outcome (state of nature). Knowing the probability distribution of outcomes (states of nature), the decision-maker needs to select a course of action resulting a largest expected (average) payoff value. The expected payoff is the sum of all possible weighted payoffs resulting from choosing a decision alternative.

The widely used criterion for evaluating decision alternatives (courses of action) under risk is the *Expected Monetary Value (EMV)* or *Expected Utility*.

1. **Expected Monetary Value (EMV) :** EMV is obtained by adding payoffs for each course of action, multiplied by the probabilities associated with each state of nature.

The expected monetary value (EMV) for a given course of action is obtained by adding payoff values multiplied by the probabilities associated with each state of nature.

The Procedure to calculate EMV

- Construct a payoff matrix listing all possible courses of action and states of nature. Enter the conditional payoff values associated with each possible combination of course of action and state of nature along with the probabilities of the occurrence of each state of nature.
 - Calculate the EMV for each course of action by multiplying the conditional payoffs by the associated probabilities and adding these weighted values for each course of action.
 - Select the course of action that yields the optimal EMV.
2. **Expected Opportunity Loss (EOL) :** Expected opportunity loss (EOL), also called *expected value of regret*, is an alternative decision criterion for decision making under risk. The EOL is defined as the difference between the highest profit (or payoff) and the actual profit due to choosing a particular course of action in a particular state of nature. Hence, EOL is the amount of payoff that is lost by not choosing a course of action resulting to the minimum payoff in a particular state of nature. A course of action resulting to the minimum EOL is preferred.
 3. **Expected Value of Perfect Information (EVPI) :** Expected value of perfect information is an average (or) value of an additional information if it were of any worth.

If decision-makers can get *perfect (complete and accurate) information* about the occurrence of various states of nature, then choosing a course of action that yields the desired payoff in the presence of any state of nature is easy.

VIII. DECISION TREE ANALYSIS

Decision tree analysis is the process of drawing a decision tree, which is a graphic representation of various alternative solutions that are available to solve a given problem, in order to determine the most effective courses of action.

Decision tree is the graphical display of the progression of decision and random events.

Decision-making problems discussed earlier were limited to arrive at a decision over a fixed period of time. That is, payoffs, states of nature, courses of action and probabilities associated with the occurrence of states of nature were not subject to change.

A decision tree analysis involves the construction of a diagram that shows, at a glance, when decisions are expected to be made – in what sequence, their possible outcomes, and the corresponding payoffs. A decision tree consists of *nodes, branches, probability estimates, and payoffs*.

There are two types of nodes:

- **Decision (or act) node:** A decision node is represented by a square and represents a point of time where a decision-maker must select one *alternative course of action* among the available. The courses of action are shown as *branches* or *arcs* emerging out of decision node.

• **Chance (or event) node:** Each course of action may result in a *chance node*. The chance node is represented by a circle and indicates a point of time where the decision-maker will discover the response to his decision.

Branches emerge from and connect various nodes and represent either decisions or states of nature.

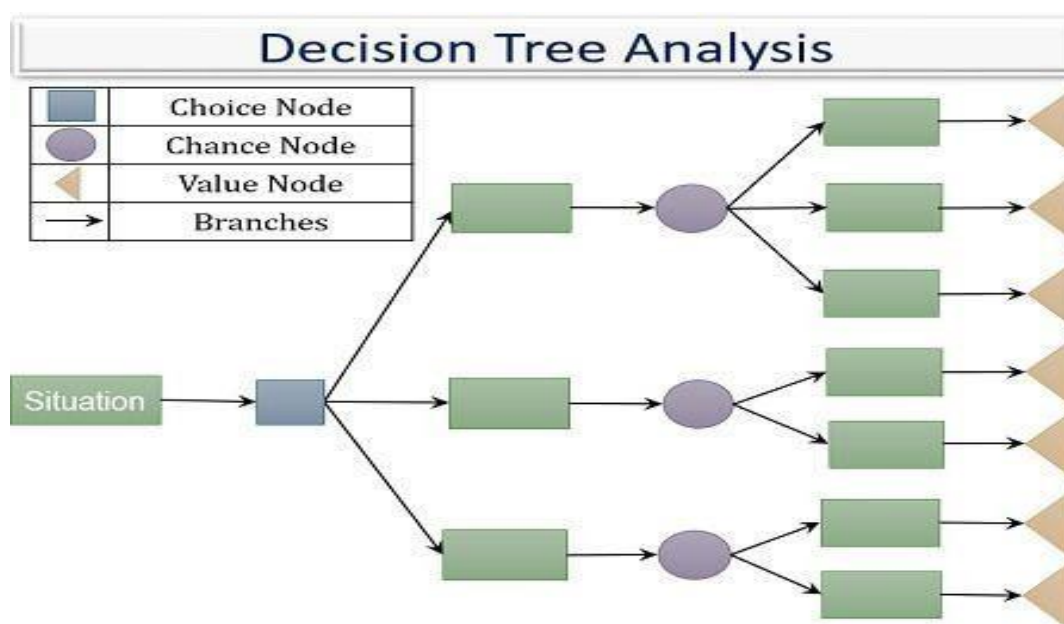
There are two types of branches:

• **Decision branch:** It is the branch leading away from a decision node and represents a course of action that can be chosen at a decision point.

• **Chance branch:** It is the branch leading away from a chance node and represents the state of nature of a set of chance events. The assumed probabilities of the states of nature are written alongside their respective chance branch.

• **Terminal branch:** Any branch that makes the end of the decision tree (not followed by either a decision or chance node), is called a *terminal branch*. A terminal branch can represent either a course of action. The terminal points of a decision tree are supposed to be mutually exclusive points so that exactly one course of action will be chosen.

The *payoff* can be positive (i.e. revenue or sales) or negative (i.e. expenditure or cost) and it can be associated either with decision or chance branches.



Remark *Decision trees versus probability trees* Decision trees are basically an extension of probability trees. However, there are several basic differences:

- The decision tree utilizes the concept of ‘rollback’ to solve a problem. This means that it starts at the right-hand terminus with the highest expected value of the tree and works back to the current or beginning decision point in order to determine the decision or decisions that should be made. It is the multiplicity of decision points that make the rollback process necessary.
- The probability tree is primarily concerned with calculating the probabilities, whereas the decision tree utilizes probability factors as a means of arriving at a final answer.
- The most important feature of the decision tree, is that it takes time differences of future earnings into account. At any stage of the decision tree, it may be necessary to weigh differences in immediate cost or revenue against differences in value at the next stage.

JOB SEQUENCING

I. MEANING : **Sequencing problem** is the problem of finding an optimal sequence of completing certain number of jobs so as to minimize the total elapsed time between completion of first and last job.

The optimal order (sequence) shows the minimum time in which jobs, equipment, people, materials, facilities and all other resources are arranged to support the production schedules to give low costs and high utilizations. Other objectives of calculating optimal production schedule are minimizing customers waiting time for a product or service, meeting promised delivery dates, keeping stock levels low providing preferred working pattern, and so on.

II. NOTATIONS, TERMINOLOGY AND ASSUMPTIONS

1. NOTATIONS

T_{ij} = Processing time (time required) for job i on machine j .

T = Total elapsed time for processing all the jobs. This includes idle time, if any.

I_{ij} = Idle time on machine j from the end of job $(i - 1)$ to the start of job i .

2. TERMINOLOGY

- **Number of Machines** ; The number of machines refer to the number of service facilities through which a job must pass before it is assumed to be completed.
- **Processing Time** : This is the time required by a job on each machine.
- **Processing Order** : This refers to the order (sequence) in which machines are required for completing the job.
- **Idle Time on a Machine** : This is the time during which a machine does not have a job to process, i.e. idle time from the end of job $(i - 1)$ to the start of job i .
- **Total Elapsed Time** : This is the time interval between starting the first job and completing the last job, including the idle time (if any), in a particular order by the given set of machines.
- **No Passing Rule** : This refers to the rule of maintaining the order in which jobs are to be processed on given machines. For example, if n jobs are to be processed on two machines, M_1 and M_2 in the order M_1, M_2 , then each job should go first to machine M_1 and then to M_2 .

3. ASSUMPTIONS

- The processing time on different machines are exactly known and are independent of the order of the jobs in which they are to be processed.
- The time taken by the job in moving from one machine to another is negligible.
- Once a job has begun on a machine, it must be completed before another job can begin on the same machine.
- All jobs are known and are ready for processing before the period under consideration begins.
- Only one job can be processed on a given machine at a time.
- Machines to be used are of different types.
- The orders of completion of jobs are independent of the sequence of jobs.

4. IMPORTANCE OF JOB SEQUENCING

A. Improved Efficiency and Utilization:

- **Reduced Idle Time:** By carefully sequencing jobs, you can minimize the time machines or workers spend idle waiting for the next job.
- **Optimized Flow:** Sequencing helps ensure a smooth flow of work through the production process, reducing bottlenecks and delays.
- **Increased Machine Utilization:** A well-sequenced workload maximizes the use of available resources, preventing them from sitting idle and underutilized.

B. Cost Reduction:

- **Lower Processing Time:** Sequencing can reduce the overall time required to complete all jobs, leading to faster turnaround times and potentially lower costs.
- **Reduced Inventory Costs:** By optimizing job flow and minimizing idle time, you can reduce the need for large inventories of work-in-process.
- **Lower Labor Costs:** Efficient scheduling can lead to better labor utilization and reduce the need for overtime or extra staff.

C. Customer Satisfaction and Delivery Reliability:

- **Meeting Deadlines:** Sequencing helps prioritize jobs based on their due dates, ensuring that customers' needs are met and deadlines are achieved.
- **Reduced Delays:** By minimizing delays and optimizing the workflow, you can improve customer satisfaction and build trust in your delivery reliability.

D. Enhanced Profitability:

- **Higher Throughput:** Increased efficiency and reduced lead times translate to higher production output and potentially greater sales.
- **Improved Profit Margins:** By reducing costs and improving efficiency, you can increase your profit margins and overall profitability.